NGUYEN CONG LAP NHAN

Developer & AI Engineer

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Ho Chi Minh City, Vietnam

Education

B.Sc. in Information Technology

2017 - 2021

University of Science

Advanced Research in Blockchain & Financial Technology (FinTech)

- In-depth research has been conducted on DeFi protocols, Web3 architecture, and smart contract logic.
- Applying quantitative methods to analyze the microstructure and token economics of the cryptocurrency market.
- I have completed intensive training in digital content creation, focusing on community building and technical storytelling for technology products.
- Participate in reporting and processing statistics for large-scale marketing/advertising campaigns

Career goals

- **Short-term:** Software Engineer / QA Automation with a quantitative mindset, using my programming strengths to **control and transform AI into a powerful assistant** in product testing and development processes. In the future, I aspire to leverage my expertise in Python (Data/AI), Node.js (Backend/System), and React (Frontend) to develop high-performance financial technology products, solve big data problems, and optimize user experience.

- **Long-term:** Become an **AI-Assisted Full-stack Developer / SaaS Engineer (Software-as-a-Service)**. Apply sharp thinking and master next-generation AI tools (such as **OpenClaw AI**) to accelerate product development timescale (SDLC), optimize code generation performance, and contribute to building large-scale SaaS systems.

- I aspire to apply my logical thinking and programming skills to solve complex data problems, while also contributing responsibly to the overall development of the engineering team at the company.

Work experience

Full-stack Developer / Data Engineer **06/2021 - 07/2023**

Softworld Vietnam

Objective: A web application that connects and sends anonymous/public messages with integrated security authentication.

Web Application Developer (Node.js):

- **Backend**: Building a RESTful API with Node.js (Express), handling user authentication via JWT and Supabase Auth.
- **Database**: Design and manage a database on Supabase to store messages and user profiles.
- As a **Data Engineer**, design and optimize data pipelines to securely collect, process, and store user data on PostgreSQL/Supabase.

Features:

- Integrates a **Resend** service for sending automatic notification emails (Transactional Email). Features a responsive interface and animations (Floating Hearts) using Vanilla JS/CSS.
- Write automated user interface (**e2e**) testing scripts using **Playwright** to ensure smooth interaction of functionalities (such as real-time messaging and JWT authentication).

Customer-centric thinking:

- Actively listen to feedback to improve the interface. Apply empathy to design a user-friendly experience (UX), helping users feel truly connected.

Blockchain Data Engineer / Researcher **08/2023 - 06/2024**

Research & Experimentation in Blockchain & Crypto

('Happy New Year' campaign, 'The Whale Hunt' campaign).

Skills

AI Coding Agents & Tools:

- Proficient in using OpenClaw AI and Bolt.new to set up workspace memory, generate automatic code, optimize performance, and debug Frontend/Backend systems.

Programming languages:

- Python (Advanced - Data processing in-depth), JavaScript/Node.js (Proficient), SQL.

Data Science & Quantitative Analysis:

- **Processing & Analysis:** Pandas, NumPy, Technical Analysis (RSI, EMA, Bollinger Bands, ADX).

Software Testing & Quality Assurance (QA/QC):

- Master software **testing knowledge** and quality assurance thinking.
- Ability to analyze requirements, create **test plans**, and write **test cases** covering complex business flows.
- Proficient in Automation Testing: Building **e2e** test scenarios with **Playwright**, performance testing and **API testing** with K6, Python (Vectorbt, PyTest).

Machine Learning & Deep Learning:

- Proficient in **machine learning** and core libraries such as **scikit-learn**, XGBoost, and LightGBM for financial trend forecasting.
- Possesses a foundation in researching and integrating computer vision models to automatically identify and classify chart patterns in trading.

System & Backend Programming:

- **Technologies:** Node.js, Express.js, Asynchronous programming (Async/Await).

Objective: To build a dashboard for monitoring real-time on-chain data.

- **Frontend & Visualization**: Using React.js and the Recharts library to plot a Netflow chart, a Paymaster activity Heatmap, and a real-time Gas chart.
- **Blockchain Data**: Using Ethers.js and RPC Nodes to scan blocks, parse transaction data, and identify Account Abstraction (AA) and Paymaster transactions.
- **On-chain Data Interaction**: Managed complex digital asset portfolios across multiple chains using non-custodial wallets (MetaMask, Trust Wallet), and researched methods to automate transaction signing via code.
- **Community Support & Empathy**: Directly listening to the difficulties of new users, and then building easy-to-understand tutorials and providing thorough answers to their questions, helping them feel confident in participating in the market.

Quantitative Developer / AI Engineer & Data Engineer
07/2024 - 07/2025

Objective: To develop and operate a comprehensive automated trading ecosystem, achieving high accuracy in trade execution and effective risk management.

Developing an automated trading system:

- A powerful trading bot has been designed, integrated with the API, featuring real-time status management, automatic reconnection logic, and advanced order execution.
- Implemented the "Smart Hunter" algorithm combined with automatic stop-loss orders, flexible position sizing based on account balance, and multi-timeframe trend analysis.
- Integrating AI to generate and process technical indicators and funding rates for "flash investment" decision-making.

Machine learning and predictive modeling (Python):

- A comprehensive ML process has been developed, utilizing combined learning methods (LightGBM combined voting classifier, XGBoost, Random Forest, and MLP) to predict market volatility.
- Advanced financial features have been designed, including Volatility Adjusted Return, Efficiency Ratio, and Z-Score, to improve the model's accuracy.
- Apply the "Step-by-Step Testing" method to prevent overfitting and ensure the reliability of the model under real-world market conditions.

Automated Testing & Quality Management:

- Build **test plans** and thousands of **test cases** (via vectorbt) to test algorithms on years of historical data.
- Use AI Agents to support the automatic generation of edge test scenarios.
- Establish a continuous **API testing** process to evaluate the latency and load capacity of third-party integration systems.

- **Protocols:** RESTful API, WebSocket (Real-time market data processing).
- **Systems:** Deployment on Windows, State Management, Automated error handling.

Frontend & Web3 Programming:

- **Web:** React.js (Hooks), Data Visualization (Recharts - Financial Charting), HTML5/CSS3.
- **Blockchain:** Ethers.js, ZkSync Era Network Interaction, Account Abstraction (Paymaster/AA Transaction Detection).

Artificial Intelligence (AI/LLM):

- Integrated Google Gemini API, Prompt Engineering (Optimizing statements for financial sentiment analysis).

Database & Cloud:

- PostgreSQL (Query optimization), Supabase (Authorization & Database authentication).

Foreign Languages:

- Excellent English (Proficient in communicating, reporting, and writing technical documents entirely in English).

Soft Skills:

- **Logical & Critical Thinking**: System problem analysis, algorithm optimization based on data.
- **Communication & Empathy**: Ability to actively listen to understand the needs of clients/colleagues. Empathy to thoroughly solve problems.
- **Responsibility**: High commitment to project deadlines and product quality (zero critical bugs).

Interests

- **Algorithmic Trading Research:** Continuously experimenting with Mean Reversion and Trend Following

- Rigorous parameter optimization has been implemented to maximize the Sharpe Rate and minimize the Maximum Dropout Rate.
- Automate the generation of performance reports (Win Rate, Profit/Loss, Equity Curve) to validate trading hypotheses before actual implementation.

Optimization And Improvement **08/2025 - 12/2025**

Objective: To develop a hybrid trading system combining technical rules and AI on the CEX exchange. The required trading algorithm must be robust enough to manage large capital with minimal losses.

System Architecture (Core Engine):

- Building a high-performance trading bot using Node.js with an asynchronous (Async/Await) architecture. Processing real-time WebSocket data streams from the Bybit v5 API, integrating auto-reconnect and state persistence mechanisms, ensuring 99.9% uptime.

AI & Machine Learning Integration:

- Combining **LLM APIs** to analyze market sentiment and filter signal noise.
- Implementing an **Ensemble Learning** model (combining LightGBM, XGBoost) and creating advanced financial features (Z-Score, Volatility-Scaled Returns) to predict price trends with significantly improved accuracy.
- This research applies **deep learning** using the **TensorFlow/PyTorch** framework to predict time series data. It also tests **computer vision** algorithms to automatically scan and classify candlestick patterns (price action patterns) from visual charts.

Advanced Testing & QA System:

- Developed a simulation framework using **Python (vectorbt, pandas)** on a 2-year historical tick dataset. Optimized parameters (Grid Search) for EMA, RSI, and ATR indicators to maximize **Sharpe Ratio** and minimize account drawdown.
- Applied **software testing knowledge** to develop an automated framework. Used **e2e testing** to simulate the entire trading journey from signal reception to profit taking.
- Built-in **API test** scenarios using K6 for real-time WebSocket data streams (Bybit v5 API).
- Applied Walk-forward testing, equivalent to advanced integration testing.

Automated Risk Management:

- Programming dynamic capital management logic based on Kelly Criterion and intelligent Trailing Stop mechanism based on volatility (ATR), ensuring the elimination of look-ahead bias during operation.

Main Role:

- Providing technical solution direction, taking primary responsibility for operational performance and risk control of the entire trading system.

HeartConnect (SaaS Web Application) **01/2026 - 03/2026**

Full-stack Developer

strategies using Python and ML libraries.

- **Knowledge Sharing:** Blogging about technical insights, algorithmic trading tutorials, and blockchain innovations to empower the developer community.
- **Community Building:** Engaging with and growing social platforms focused on technology and financial literacy.
- **Digital Marketing and Content Strategy**: Completed intensive training in digital content creation, focusing on community building and technical storytelling for technology products.

Independently develop the entire system from Front-end (React, Tailwind CSS) to Back-end (Node.js, Supabase, Resend).

- Using **OpenClaw AI** as the primary programming assistant: Manage context through workspace memory (.openclaw/workspace/memory), generate code automatically, and definitively handle complex logic errors (such as routing/dashboard errors).
- I complete the entire process, from ideation and documentation writing (README.md) to product development.

Titles and awards

2023 - 2025

- **Algorithmic Trading Optimization:** Achieved a consistently profitable strategy (Sharpe Ratio > 1.7 in backtests) by applying "Walk-Forward Validation" to eliminate overfitting in trading models.

Certificate

2024

Quantitative Finance & Data Analysis Mastery (Self-Study Equivalent to Professional Standards)

Focus: Financial time series analysis, Machine learning in trading, and Risk management systems.

2025

Core Competency: Proficient in translating complex financial theories (Smart Money Concepts, Elliott Wave) into executable code (Python/Node.js).

More information

- **Tech Stack:** Strong command of Python (Data Science stack: Pandas, Numpy, Scikit-learn) and JavaScript (Node.js backend logic).
- **Integrating GenAI:** Experience using LLM for financial psychology analysis and programming.
- **Mindset:** Data-driven, disciplined in risk management, and constantly updating with the latest developments in AI and Web3 infrastructure.

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